

Titel: Analysis of Basic Network Security Mechanisms and Traffic Monitoring

Abstract : This report presents an analysis of basic network security mechanisms and traffic monitoring techniques. The objective is to understand common network threats, such as viruses, worms, trojans, and phishing attacks, and explore how security tools like firewalls and encryption can help protect network environments. A simple network setup was configured to implement essential security measures, including enabling a firewall, changing default credentials, and using WPA2/WPA3 encryption. Network traffic was then monitored and analyzed using Wireshark to identify normal and suspicious activities. The findings highlight the importance of proactive security configurations and continuous traffic monitoring to maintain a secure and reliable network infrastructure.

Theory: Network security is the practice of protecting a computer network and its data from unauthorized access, misuse, or theft. It involves a combination of hardware, software, policies, and procedures designed to defend network systems from internal and external threats. The main goal of network security is to ensure confidentiality, integrity, and availability of data often referred to as the CIA triad.

To understand network security, it is important to study the different types of network threats.

- Viruses are malicious programs that attach themselves to legitimate files and spread across systems, damaging or altering data.
- Worms are self-replicating malware that exploit vulnerabilities in software to spread without human interaction.
- Trojans disguise themselves as useful applications but execute harmful actions once installed.
- Phishing attacks use deceptive emails or websites to trick users into revealing personal or financial information.

To counter these threats, several security mechanisms are used:

- Firewalls act as a barrier between trusted and untrusted networks, filtering incoming and outgoing traffic based on security rules.
- Encryption ensures that data transmitted across the network is unreadable to unauthorized users.
- Secure configurations, such as changing default passwords and enabling WPA2/WPA3 encryption on wireless networks, prevent unauthorized access.
- Network monitoring tools, like Wireshark, help in capturing and analyzing packets to detect suspicious or malicious traffic patterns.

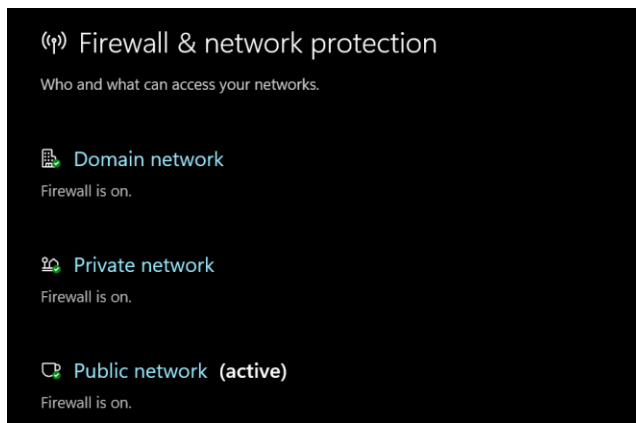
By implementing these mechanisms, network administrators can identify threats early, limit potential damage, and maintain the overall security of network operations. Traffic monitoring plays a key role in detecting anomalies and ensuring that communication within the network remains safe and efficient.

Security Measures Implemented: To strengthen the network against common security threats, several basic yet effective security measures were implemented. These measures ensure that the network remains protected, monitored, and resilient against unauthorized access or attacks. The implementation focused on three major aspects: prevention, detection, and protection.

1. Firewall Configuration: A firewall acts as the first line of defense in a network by filtering incoming and outgoing traffic based on predefined security rules.

- The Windows Defender Firewall was enabled and configured to block unauthorized inbound connections while allowing only necessary outbound connections.
- Custom rules were created to restrict access to specific ports and applications, minimizing the risk of intrusion.
- Logs were monitored to track blocked traffic and identify any suspicious connection attempts.

This configuration ensures that only legitimate traffic can enter or leave the network, reducing the risk of external attacks.



2. Network Encryption (WPA2/WPA3):

To secure wireless communications, WPA2 encryption was enabled on the local Wi-Fi network.

- This ensures that data transmitted between devices is encrypted and cannot be easily intercepted by attackers.
- For networks supporting modern standards, WPA3 can be implemented to provide enhanced protection against brute-force and dictionary attacks.
- Strong, unique passwords were used to prevent unauthorized access to the network.

Encryption helps maintain the confidentiality and integrity of data during transmission.

3. Changing Default Credentials:

One of the simplest yet most effective steps in securing a network is changing default router and device passwords.

- The router's default administrator credentials were updated with a strong password containing a mix of uppercase, lowercase, numbers, and special characters.
- Similarly, any connected devices (such as PCs or IoT devices) were assigned unique and secure login credentials.

This prevents unauthorized users from exploiting factory-set credentials to gain control over the network.

4. System and Software Updates:

Keeping all devices up to date is a critical part of network security.

- The operating system and antivirus software were regularly updated to patch known vulnerabilities.
- Automatic updates were enabled to ensure timely installation of security patches and improvements.

This step reduces the chances of attackers exploiting outdated software or unpatched vulnerabilities.

5. Wireshark Network Monitoring:

The Wireshark tool was used to capture and analyze network traffic to detect unusual or malicious patterns.

- Traffic was monitored for HTTP, DNS, and TCP packets to understand normal communication behavior.
- Suspicious packets or unusual traffic spikes were identified for further analysis.
- The tool helped visualize packet flow, revealing potential signs of intrusion or unauthorized access.

Wireshark provided valuable insights into the real-time behavior of the network, supporting proactive security management.

6. Secure Network Configuration:

Additional configurations were applied to enhance overall network security:

- Disabling unused network services and ports to minimize the attack surface.
- Enabling network discovery control to limit visibility to trusted devices only.
- Configuring router firewalls and access control lists (ACLs) to restrict unauthorized device access.

These configurations collectively strengthen the network's defense against both internal and external threats

Wireshark Traffic Observations

To analyze network behavior and identify potential security issues, **Wireshark** was used to capture and study live network traffic. Wireshark allows packet-level inspection, providing detailed insights into communication protocols, data exchange, and possible vulnerabilities within the network.

1. Overview of Network Capture:

Network traffic was captured for a period of several minutes while performing regular internet activities such as web browsing and domain lookups.

During the analysis, several key protocols were observed:

- **HTTP (Hypertext Transfer Protocol):** Used for unencrypted web communication.
- **DNS (Domain Name System):** Used to translate domain names into IP addresses.

• 2. HTTP Traffic Analysis:

File Machine View Input Devices Help

Apps Places

Oct 28 16:12

Capturing from eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
733	134.205904942	44.228.249.3	10.247.99.225	HTTP	785	HTTP/1.1 504 Gateway Time-out (text/html)
1186	220.259550901	44.228.249.3	10.247.99.225	HTTP	785	HTTP/1.1 504 Gateway Time-out (text/html)
420	74.562408408	10.247.99.225	44.228.249.3	HTTP	762	POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded)
925	159.774605604	10.247.99.225	44.228.249.3	HTTP	762	POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded)

Frame 420: 762 bytes on wire (6096 bits), 762 bytes captured (6096 bits) on interface eth0, id 0

Ethernet II, Src: AzureWaveTec_c4:06:67 (10:68:38:c4:06:67), Dst: 42:6f:eb:46:9e:36 (42:6f:eb:46:9e:36)

Internet Protocol Version 4, Src: 10.247.99.225, Dst: 44.228.249.3

Transmission Control Protocol, Src Port: 60233, Dst Port: 80, Seq: 1, Ack: 1, Len: 708

Hypertext Transfer Protocol

HTML Form URL Encoded: application/x-www-form-urlencoded

Form item: "uname" = "kirtaneanikt@gmail.com"

Key: uname

Value: kirtaneanikt@gmail.com

Form item: "pass" = "A@1N#2k\$3i4T"

Key: pass

Value: A@1N#2k\$3i4T

Hypertext Transfer Protocol: Protocol

Packets: 1973 · Displayed: 4 (0.2%)

The captured **HTTP traffic** showed communication between the local host and external web servers. The analysis revealed:

- **Request and Response Headers:** Containing information such as user-agent, host, and content type.
- **Unencrypted Transmission:** HTTP operates without encryption, meaning data such as form inputs and cookies can be intercepted if not secured with HTTPS.
- **Potential Risks:** Attackers on the same network could perform packet sniffing to capture sensitive information transmitted over plain HTTP.

Observation: The analysis highlighted the importance of using **HTTPS (HTTP Secure)** instead of HTTP to ensure that all data exchanged between the client and server is encrypted and protected from eavesdropping.

3. DNS Traffic Analysis:

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
477	74.49240667	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	211	Standard query response 0x7ec9 A edge-microsoft.com CNAME edge-microsoft.com ax-0002.ax-msedge.net CNAME ax-0002.ax-msedge.net
482	75.47977870	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	226	Standard query response 0x4967 HTTPS edge-microsoft.com CNAME edge-microsoft.com ax-0002.ax-msedge.net CNAME ax-0002.ax-msedge.net
822	139.958591951	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	304	Standard query response 0xbdf0 HTTPS functional.events.data.microsoft.com CNAME global.asimov.events.data.trafficmanager.net CNAME
823	139.958592051	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	304	Standard query response 0xbdf0 AAAA functional.events.data.microsoft.com CNAME global.asimov.events.data.trafficmanager.net CNAME
825	139.958592242	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	256	Standard query response 0x9335 A functional.events.data.microsoft.com CNAME global.asimov.events.data.trafficmanager.net CNAME
874	160.578915978	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	366	Standard query response 0x62b6 AAAA www.bing.com CNAME www.bing.com trafficmanager.net CNAME www.bing.com edgekey.net CNAME
875	160.578916419	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	370	Standard query response 0x7c59 A www.bing.com CNAME www.bing.com trafficmanager.net CNAME www.bing.com edgekey.net CNAME
876	160.578916529	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	287	Standard query response 0x7752 HTTPS www.bing.com CNAME www.bing.com trafficmanager.net CNAME www.bing.com edgekey.net CNAME
877	160.57891712	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	171	Standard query response 0xb2b6 AAAA testphp.vulnweb.com SOA ns1.eurodns.com
878	160.627485097	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	287	Standard query response 0xe095 AAAA testphp.vulnweb.com SOA ns1.eurodns.com
1127	208.926763524	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	241	Standard query response 0x5413 A watson.events.data.microsoft.com CNAME blobcollectorcommon.trafficmanager.net CNAME onedsblob
1129	208.926859817	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	283	Standard query response 0xb79c AAAA watson.events.data.microsoft.com CNAME blobcollectorcommon.trafficmanager.net CNAME onedsblob
1271	271.096456795	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	287	Standard query response 0x3d55 HTTPS www.bing.com CNAME www.bing.com trafficmanager.net CNAME www.bing.com edgekey.net CNAME
1272	271.096457617	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	370	Standard query response 0x6877 A www.bing.com CNAME www.bing.com trafficmanager.net CNAME www.bing.com edgekey.net CNAME
1392	338.269954285	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	282	Standard query response 0x5167 AAAA watson.events.data.microsoft.com CNAME blobcollectorcommon.trafficmanager.net CNAME onedsblob
1393	338.269955177	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	241	Standard query response 0xf9ae A watson.events.data.microsoft.com CNAME blobcollectorcommon.trafficmanager.net CNAME onedsblob
1587	407.097346099	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	228	Standard query response 0x2a5e AAAA watson.events.data.microsoft.com CNAME blobcollectorcommon.trafficmanager.net CNAME onedsblob
1588	407.097347571	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	244	Standard query response 0x7f93 A watson.events.data.microsoft.com CNAME blobcollectorcommon.trafficmanager.net CNAME onedsblob
1909	478.778774896	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	237	Standard query response 0x6813 AAAA fp.msedge.net CNAME 1.perf.msedge.net CNAME a-0019.a.msedge.net CNAME a-0019.a.msedge.net
1910	478.778775738	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	253	Standard query response 0x1283 A fp.msedge.net CNAME 1.perf.msedge.net CNAME a-0019.a.msedge.net CNAME a-0019.a.msedge.net
1963	481.447163642	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	379	Standard query response 0xd94a A login.live.com CNAME login.msa.asiaidentity.com CNAME www.msft-id-prod.admsa.akadns.net CNAME pr
1966	481.538834273	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	389	Standard query response 0x7144 AAAA login.live.com CNAME login.msa.asiaidentity.com CNAME www.msft-id-prod.admsa.akadns.net CNAME pr
2014	483.128895962	2401:4900:52f7:560d::	2401:4900:52f7:560d::	DNS	298	Standard query response 0x1931 AAAA bx-rim.msedge.net CNAME bx-rim-bx-9999.bx.msedge.net CNAME bx-9999.bx.msedge.net AAAA 2003

Domain Name System (response)

Length: 295

Transaction ID: 0x7c60

Flags: 0x0180 Standard query response, No error

Questions: 1

Answer RRs: 12

Authority RRs: 0

Additional RRs: 0

Queries

Answers

www.bing.com type CNAME, class IN, cname www.bing.com.trafficmanager.net

www.bing.com type CNAME, class IN, cname www.bing.com.trafficmanager.net

www.bing.com.edgekey.net type CNAME, class IN, cname e86303.dsxc.akamaiedge.net

e86303.dsxc.akamaiedge.net type A, class IN, addr 23.46.9.93

e86303.dsxc.akamaiedge.net type A, class IN, addr 23.46.9.90

e86303.dsxc.akamaiedge.net type A, class IN, addr 23.46.9.72

e86303.dsxc.akamaiedge.net type A, class IN, addr 23.46.9.96

e86303.dsxc.akamaiedge.net type A, class IN, addr 23.46.9.73

e86303.dsxc.akamaiedge.net type A, class IN, addr 23.46.9.83

e86303.dsxc.akamaiedge.net type A, class IN, addr 23.46.9.88

e86303.dsxc.akamaiedge.net type A, class IN, addr 23.46.9.74

e86303.dsxc.akamaiedge.net type A, class IN, addr 23.46.9.112

Request in: 854

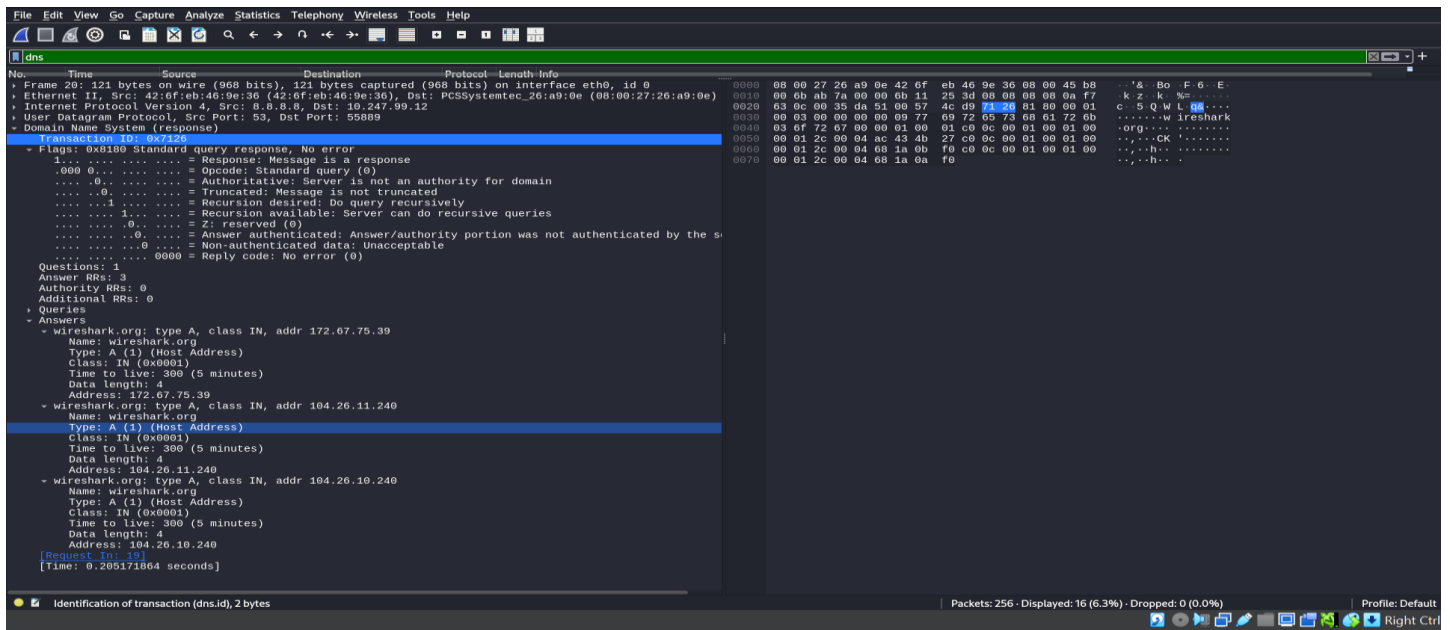
[Time: 0.219250199 seconds]

Text Item (text), 16 bytes

Frame (370 bytes) Reassembled TCP (297 bytes)

Packets: 3023 · Displayed: 217 (7.2%) · Dropped: 0 (0.0%)

Profile: Default



The DNS packets captured showed multiple queries and responses exchanged between the client device and the DNS server. To manually verify DNS resolution, the following command was executed in the terminal:

This command queries the Google Public DNS server (8.8.8.8) for the IP address of the domain `wireshark.org`. The result, displayed in the screenshot above, returned both IPv4 and IPv6 addresses for the domain, confirming successful DNS resolution despite initial timeout messages.

Observation from Capture:

- The UDP port 53 was primarily used for DNS queries and responses.
- The non-authoritative answer from the DNS server indicated external resolution from Google DNS rather than a local cache.

- The query and response process was successfully captured and verified using Wireshark, showing packet exchanges between the client and DNS server.
- Initial timeout messages indicate a slight delay in establishing communication with the DNS resolver, which may occur due to firewall filtering or temporary connectivity issues.

Security Analysis:

DNS traffic in this setup was unencrypted, making it susceptible to interception or spoofing. Attackers could exploit this vulnerability through DNS cache poisoning or man-in-the-middle attacks to redirect traffic to malicious sites. To mitigate such risks, secure DNS protocols like DNS over HTTPS (DoH) or DNS over TLS (DoT) should be implemented to encrypt queries and responses.

4. Key Findings from Wireshark Monitoring:

- Normal network activity included web browsing, DNS lookups, and secure HTTPS connections.
- Unusual packets or abnormal traffic patterns can be detected early using Wireshark, helping prevent data breaches.
- Traffic analysis revealed the difference between secure (HTTPS, TLS) and insecure (HTTP, DNS) communication protocols.

Discussion: How Security Measures Protect the Network

- The implementation of fundamental security measures plays a crucial role in protecting network infrastructure from both internal and external threats. Each measure contributes to enhancing the confidentiality, integrity, and availability of network data the three essential pillars of cybersecurity.
- The following discussion explains how the configured security mechanisms collectively strengthen overall network defense.

1. Firewall Configuration:

A firewall serves as a network's primary defensive boundary, filtering incoming and outgoing traffic based on predefined rules.

- It blocks unauthorized access attempts from external sources while allowing legitimate communication.
- Custom inbound and outbound rules help restrict unwanted connections and prevent malicious software from transmitting data externally.
- Continuous monitoring of firewall logs assists in identifying intrusion attempts or irregular activities.

2. Encryption (WPA2/WPA3):

Encryption ensures that all communication between network devices remains confidential and secure.

- WPA2/WPA3 encryption protects data transmitted over Wi-Fi by converting it into unreadable ciphertext for unauthorized users.

- Even if an attacker intercepts the data packets, the encrypted content remains indecipherable without the correct encryption key.
- Encrypted connections help maintain both data confidentiality and user privacy.

3. Changing Default Credentials:

Changing default credentials is a simple but highly effective preventive measure.

- Default usernames and passwords are often publicly known, making them easy targets for brute-force or automated attacks.
- By setting strong, unique passwords, the risk of unauthorized login attempts and device hijacking is minimized.
- This measure strengthens authentication and access control mechanisms within the network.

4. Network Monitoring Using Wireshark:

Continuous monitoring through Wireshark provides visibility into real-time network activities.

- Administrators can identify suspicious packets, abnormal traffic patterns, or unexpected communication attempts.
- Monitoring DNS and HTTP traffic helps detect potential phishing attempts, malware activity, or unauthorized data transfers.
- Early detection allows for prompt responses before threats escalate into major incidents.

Reflection: Security Best Practices:

Through this practical exercise, the importance of implementing strong and consistent network security measures became clear. Even in a small network environment, weak configurations or neglected security updates can expose systems to significant risks.

To maintain a secure and resilient network, the following best practices should be consistently applied:

1. **Enable Multi-Layered Security:** Combine firewalls, encryption, antivirus, and intrusion detection systems to create multiple layers of protection against various types of attacks.
2. **Use Strong Authentication:** Implement strong, unique passwords and enable multi-factor authentication (MFA) wherever possible to enhance access control.
3. **Regularly Update Software and Hardware:** Keep operating systems, routers, and applications updated with the latest patches to close known vulnerabilities.
4. **Monitor Network Traffic Continuously:** Use tools like Wireshark or Snort to detect anomalies, intrusions, or unauthorized activities in real time.
5. **Educate Users on Security Awareness:** Human error remains one of the biggest vulnerabilities. Training users to recognize phishing, social engineering, and unsafe browsing practices greatly reduces risk.
6. **Backup Data Frequently:** Regular backups protect against ransomware and data corruption, ensuring quick recovery from security incidents.

7. Implement Least Privilege Access: Limit access rights to only what users need to perform their tasks, minimizing the potential impact of compromised accounts.
8. Use Secure Communication Protocols: Replace outdated or insecure protocols (like HTTP or Telnet) with secure alternatives (HTTPS, SSH, TLS).

Conclusion:

This study provided a comprehensive understanding of fundamental network security mechanisms and their practical implementation. The experiment demonstrated how tools such as Wireshark can be used to analyze real-time traffic and identify potential security threats within a network.

Implementing basic security configurations—such as enabling firewalls, applying encryption (WPA2/WPA3), changing default credentials, and performing regular updates—proved essential for safeguarding network infrastructure. Traffic analysis also revealed how unencrypted communications (like HTTP or DNS) could expose sensitive information, reinforcing the need for secure protocols such as HTTPS and DoH/DoT.

The project highlighted that effective network security is not dependent on a single tool or technology but rather on a layered approach that combines prevention, detection, and response mechanisms. By following industry best practices and maintaining constant vigilance, individuals and organizations can ensure the confidentiality, integrity, and availability of their data, creating a secure and trustworthy digital environment.